

High-Level Design (HLD) Document of Educational Module

Educational Content Module - AOFIDPS

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1. Executive Summary

Problem Statement

Students in rural, low-connectivity, and disaster-prone regions face fragmented learning. Traditional Learning Management Systems (LMS) require continuous internet connectivity to stream videos, load interactive quizzes, and track progress. When the network drops, the pedagogical cycle breaks, leading to severe educational exclusion.

Proposed Solution

The Educational Content Module of the Adaptive Offline-First Intelligent Data Pre-Fetching System (AOFIDPS) solves this by decoupling learning from live internet access. It utilizes Edge AI to predict a student's learning trajectory, aggressively pre-fetches compressed curriculum bundles (videos, texts, assessments) during brief windows of connectivity, and executes a fully functional LMS environment locally on the device. Progress is stored as deterministic events and synced to the cloud upon reconnection.

2. System Overview

2.1 Project Background

Existing educational apps like Coursera or standard Moodle instances treat offline access as an afterthought (e.g., manual "Download for Offline" buttons). These approaches fail in areas where users do not know when they will lose or regain connectivity. The AOFIDPS Education Module acts autonomously, removing the cognitive load of bandwidth management from the student.

2.2 Key Objectives

- Achieve zero-buffering playback for predicted curriculum paths.
- Enable a fully local assessment engine with secure offline grading.
- Automate the storage lifecycle of large media files without user intervention.
- Ensure deterministic progress synchronization without data loss.

2.3 Scope and Boundaries

In Scope:

- AI-driven curriculum dependency mapping.
- Background pre-fetching of compressed media bundles.
- Offline interactive quiz engine and progress tracking.
- Semantic storage eviction (LRU + TTL).

Out of Scope:

- Synchronous live video classrooms (e.g., Zoom/WebRTC integration).
- Real-time multiplayer collaborative whiteboards.

3. Architecture Principles

3.1 Design Philosophy

- **Pedagogical Autonomy:** The local device must contain enough logic to gate content, grade assessments, and unlock subsequent modules without server approval.
- **Semantic Storage Management:** Storage is the tightest constraint. The system must understand that a completed 1st-grade math video is less valuable than an upcoming 2nd-grade physics chapter.
- **Event-Sourced Progress:** Learning states are not updated as monolithic profiles, but rather appended as a stream of granular events (e.g., VideoPaused_At_12m, QuizPassed_Score_85).

3.2 Architectural Patterns

- **Directed Acyclic Graph (DAG) Navigation:** Curriculum is modeled as a DAG. Pre-fetching algorithms traverse this graph based on user velocity.
- **CQRS (Command Query Responsibility Segregation):** Local reads (viewing content) are instantaneous. Writes (submitting a quiz) are queued as commands.
- **State Machine Eviction:** Content moves through states: Predicted -> Prefetched -> Consumed -> Retained for Review -> Evicted.

4. Technology Stack

Layer	Technology	Justification
Mobile Client	Flutter, Dart, FFmpeg (WASM/Native)	Cross-platform UI. FFmpeg enables local video decoding/transcoding of highly compressed AV1/H.265 chunks.
Local Data	Isar DB (Edge NoSQL)	Ultra-fast local querying for the assessment engine and tracking micro-progress events.
Edge AI	TensorFlow Lite (TFLite)	Runs Markov Chain probability models locally to predict the next required syllabus nodes.
Cloud Backend	FastAPI (Python), PostgreSQL	Python excels at traversing and updating complex curriculum graphs and processing analytics.
Content Delivery	S3-Compatible Object Storage, Edge CDNs	Serves static, immutable, and heavily compressed media bundles globally.

5. System Architecture

5.1 High-Level Architecture Diagram

graph TD

subgraph Mobile Edge Environment

UI[Student Interface / Video Player]

LE[Local Assessment Engine]

PE[TFLite Prediction Engine]

PM[Prefetch & Download Manager]

EC[Semantic Eviction Controller]

DB[(Encrypted Isar Local DB)]

UI <--> DB

UI --> LE

LE --> DB

DB --> PE

PE --> PM

PM --> DB

DB --> EC

end

subgraph Cloud Infrastructure

API[Cloud API Gateway]

Sync[CRDT Progress Sync Service]

CG[Curriculum Graph Engine]

CDN[(Content Delivery Network)]

API --> Sync

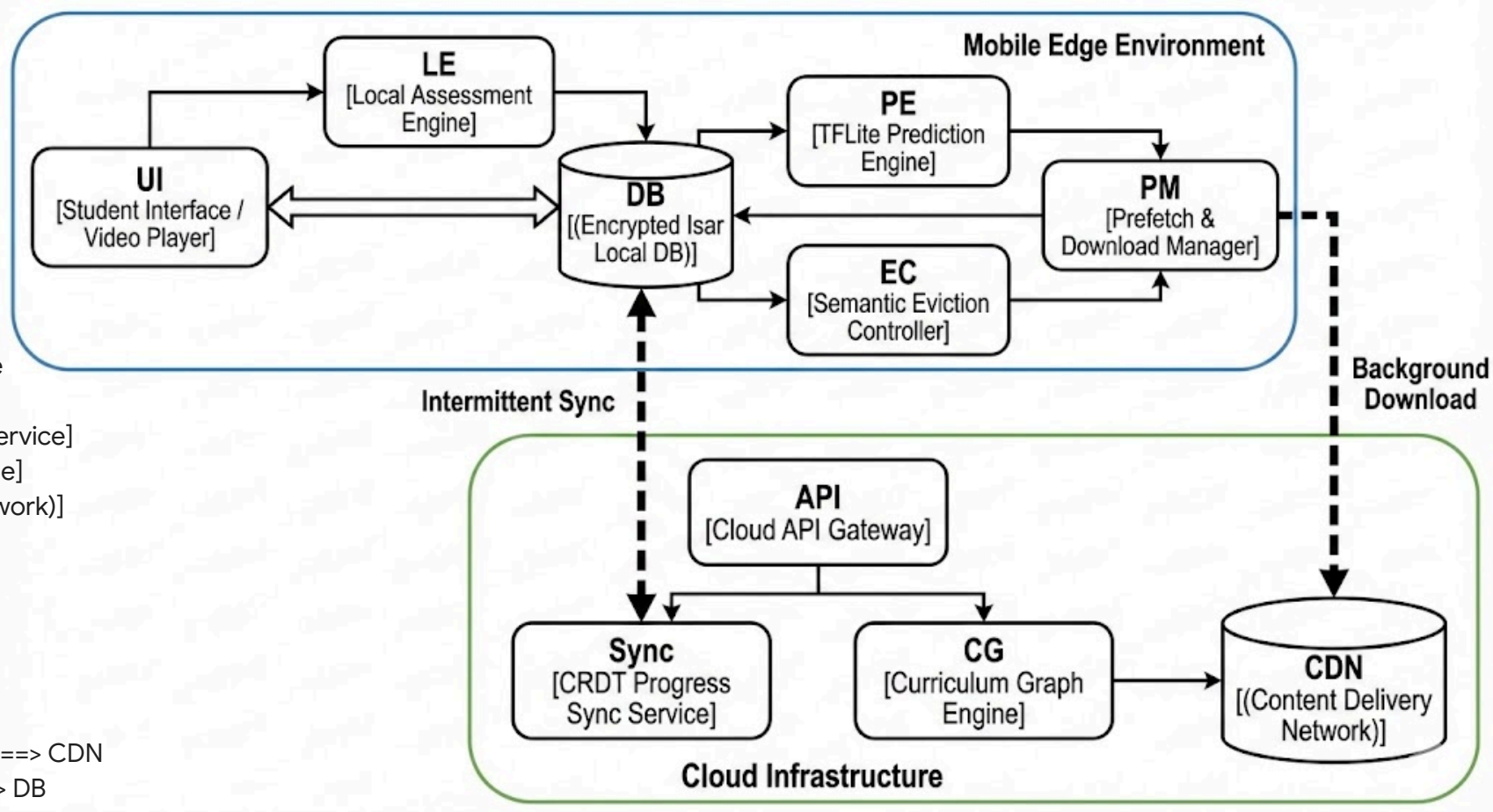
API --> CG

CG --> CDN

end

PM == Background Download ==> CDN

Sync <== Intermittent Sync ==> DB



5.2 Component Overview

- **TFLite Prediction Engine:** Analyzes current progress and learning velocity. Output: A prioritized array of Node IDs (e.g., Chapter 4, Quiz 2).
- **Prefetch Manager:** Monitors network state. When active, it pulls the predicted Node IDs from the CDN in parallel chunks.
- **Local Assessment Engine:** Parses JSON-based quiz schemas, grades answers locally, and queues the result into the Isar DB.
- **Semantic Eviction Controller:** Clears storage. It deletes raw video files of passed modules but retains lightweight text summaries for review.
- **Curriculum Graph Engine (Cloud):** Maintains the global state of course dependencies and structural prerequisites.

6. Data Architecture

6.1 Database Strategy (Optimized)

- **Mobile (Isar DB Partitions):**
 - Content_Metadata: JSON structures of chapters, text blocks, and quiz questions.
 - Media_Blobs: Pointers to encrypted local file paths for H.265 videos and WebP images.
 - Event_Queue: Un-synced user actions (e.g., {"action": "quiz_complete", "node_id": "math_101", "score": 90, "timestamp": 1715623000}).
- **Cloud (PostgreSQL):**
 - Curriculum_Nodes: Master graph of all educational content.
 - Student_Ledger: Append-only table of synced progress events.
 - Aggregated_State: Materialized view of current student grades.

6.2 Data Flow Design (Situational)

1. **Prediction:** Student completes Chapter 1 offline. Local AI predicts Chapters 2, 3, and 4 are next.
2. **Network Restored:** Device enters a 3G/Wi-Fi zone.
3. **Delta Sync:** The Event_Queue pushes Chapter 1 completion to the cloud.
4. **Prefetch:** The Prefetch Manager downloads compressed bundles for Chapters 2-4.
5. **Consumption:** User goes offline. They watch Chapter 2 videos and take the Chapter 2 quiz.
6. **Eviction:** Upon passing Chapter 2, the Eviction Controller deletes Chapter 1's heavy video files to free up space.

7. Security & Integrity Architecture

- **Anti-Cheat Validation:** Quizzes completed offline are time-stamped. The backend validates the cryptographic signature of the timestamp against time-tampering (e.g., user changing phone clock) before committing the grade to the official ledger.
- **DRM (Digital Rights Management):** Premium educational bundles are encrypted at rest on the device using AES-256. They are decrypted only in memory during playback to prevent piracy of course materials.
- **Idempotent Grading:** Event streams ensure that if a quiz completion event is synced twice due to a network drop, the backend processes the grade only once.

8. Deployment Architecture

- **Static Asset Serving:** 95% of the data volume in education is static media. This is deployed purely on Edge CDNs (e.g., Cloudflare, AWS CloudFront) localized to the target regions (e.g., Africa, Southeast Asia) to maximize download speeds during brief connectivity windows.
- **Dynamic Sync Services:** Cloud Run (Serverless containers) handle the intermittent bursts of telemetry and progress data when buses or trains of students hit Wi-Fi zones simultaneously.

9. Risk Management

Risk	Impact	Deep Mitigation Strategy
Storage Exhaustion	App crashes, unable to fetch new courses.	Implement semantic eviction. Delete videos but keep PDF transcripts. Hard-cap app storage to a percentage of OS free space.
Stale Content	Students learn outdated curriculum.	Attach TTL (Time-To-Live) tokens to metadata. If content expires, the local engine forces a delta-patch sync before allowing access.
Sync Conflicts	Student uses tablet and phone offline simultaneously.	CRDTs (Conflict-free Replicated Data Types) merge progress deterministically based on highest logical timestamp.
Bandwidth Hogging	Prefetcher drains user's limited cellular data plan.	Restrict heavy video prefetching exclusively to unmetered Wi-Fi connections, or require explicit opt-in for cellular downloads.

10. Deep Problem Analysis with Deepest Solutions

Problem 1: The "Dead End" Offline Experience

- **Deep Analysis:** Standard caching requires users to manually select "Download Module 2". If they finish Module 2 while offline, they are stuck until they find internet again.
- **Deep Solution:** Predictive Graph Traversal.
- **Implementation Approach:** The curriculum is modeled as a DAG. The Edge AI calculates the user's average velocity (e.g., 2 modules per day). The Prefetch Manager always maintains an "Offline Buffer" of *Velocity × 3 Days*. The user never hits a dead end.

Problem 2: Colossal Storage Footprint of Video Learning

- **Deep Analysis:** A standard 1080p course can be 5GB+. Low-end Android devices in rural areas often have less than 16GB of total disk space.
- **Deep Solution:** Aggressive Transcoding & Semantic Modularity.
- **Implementation Approach:** 1. Cloud-side transcoding converts all video to heavily compressed AV1/H.265 at 480p/720p maximum.
2. Audio tracks are decoupled from video. If storage is critically low (<500MB), the Eviction Controller dynamically deletes the video track but retains the audio track and PDF slides, degrading the experience gracefully rather than breaking it.

Problem 3: Offline Assessment Integrity

- **Deep Analysis:** How do you guarantee a student didn't cheat or exploit an offline quiz if the server isn't validating the answers in real-time?
- **Deep Solution:** Local Secure Enclave Grading + Heuristic Cloud Auditing.
- **Implementation Approach:** Quiz logic (correct answers) is encrypted and decrypted in memory. When the student takes the test, the Local Assessment Engine logs micro-telemetry (time spent per question, answer switching). Upon sync, the backend analyzes this telemetry array. If a student answered 50 complex calculus questions correctly in 4 seconds, the backend flags the offline test as anomalous and requires a manual retest.

11. Glossary

- **DAG:** Directed Acyclic Graph (Used for mapping curriculum prerequisites).
- **CQRS:** Command Query Responsibility Segregation.
- **CRDT:** Conflict-free Replicated Data Type.
- **Semantic Eviction:** Deleting data based on its contextual value rather than just its age.
- **LMS:** Learning Management System.